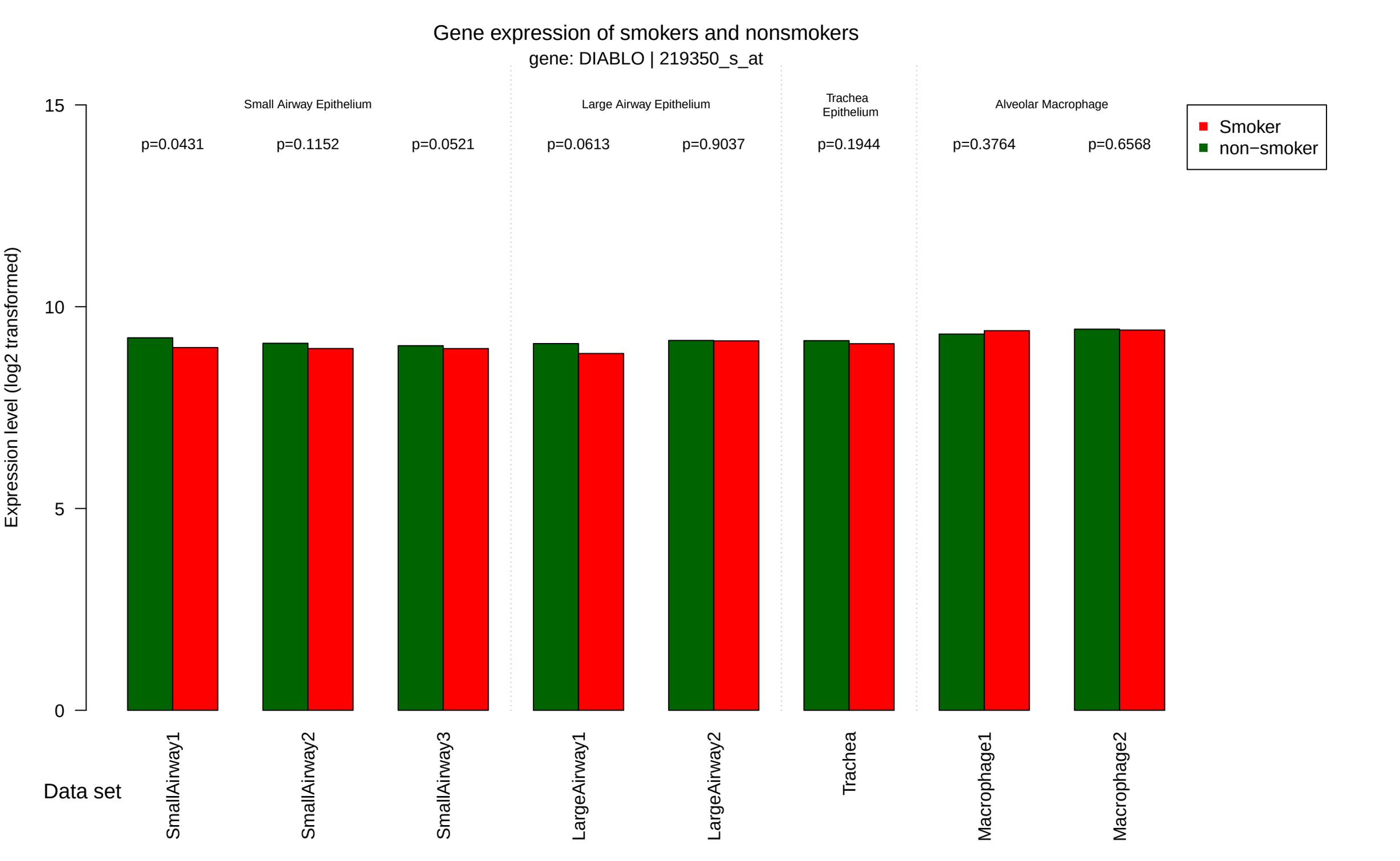
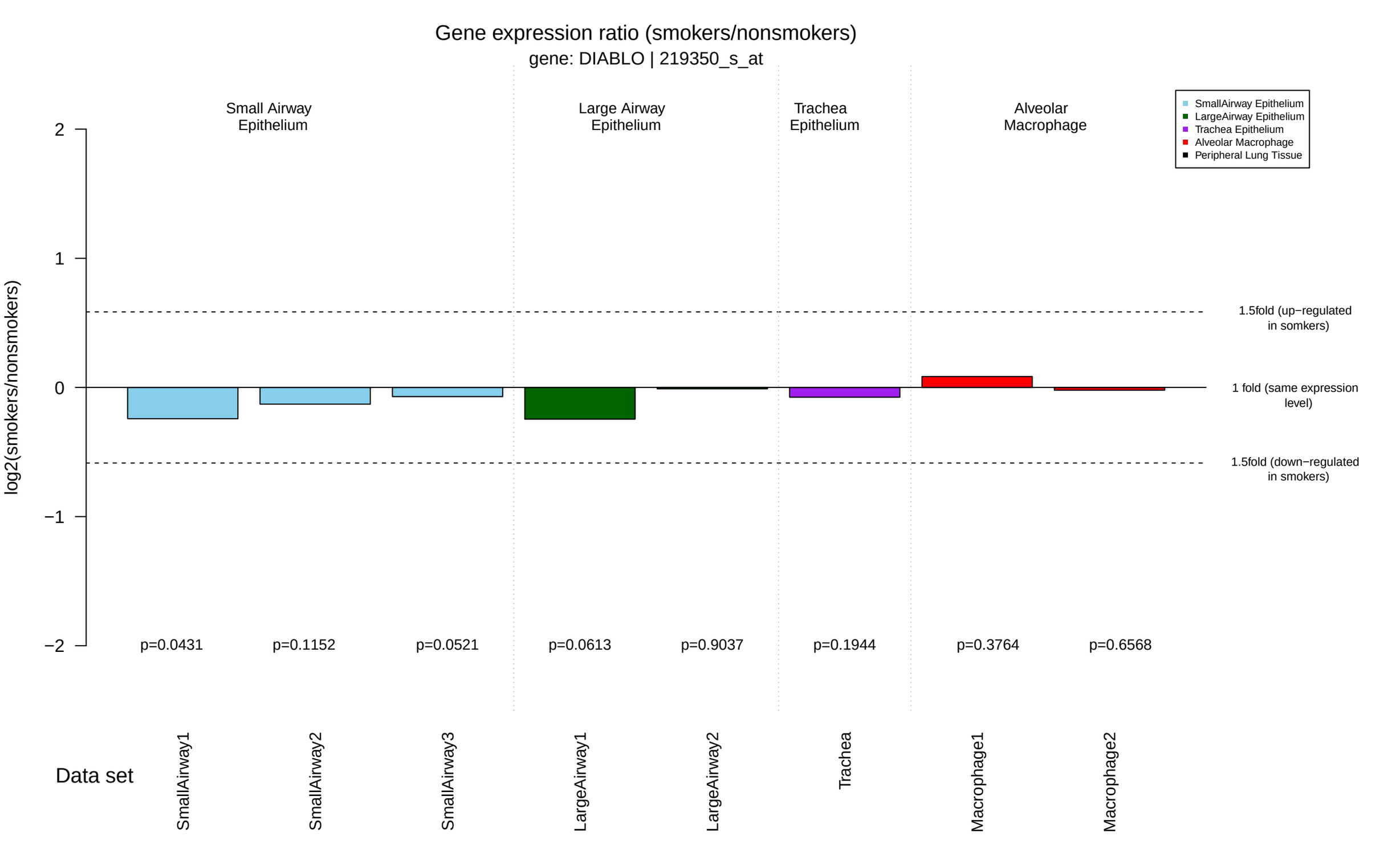
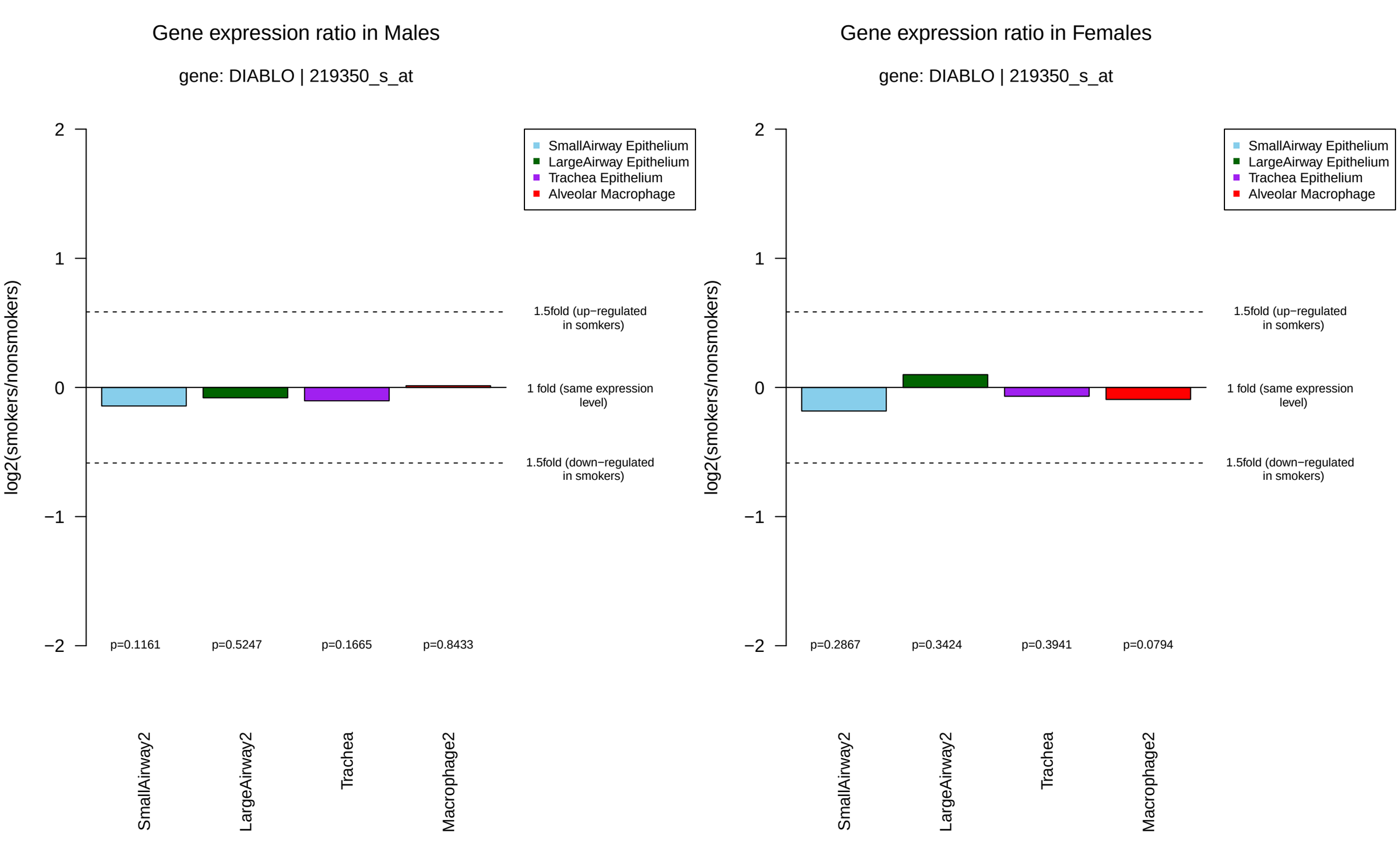
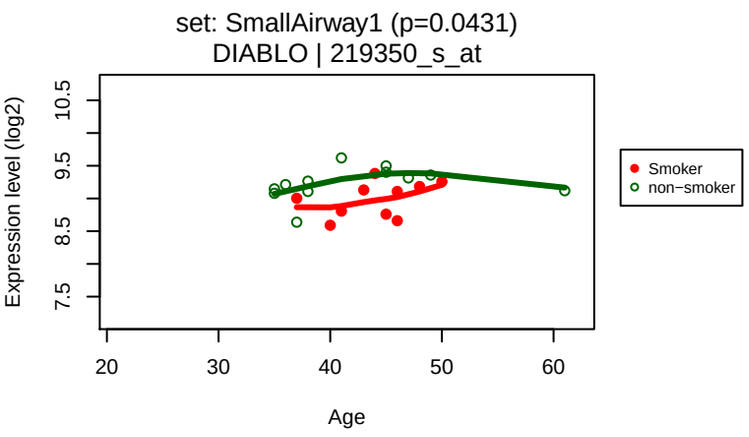
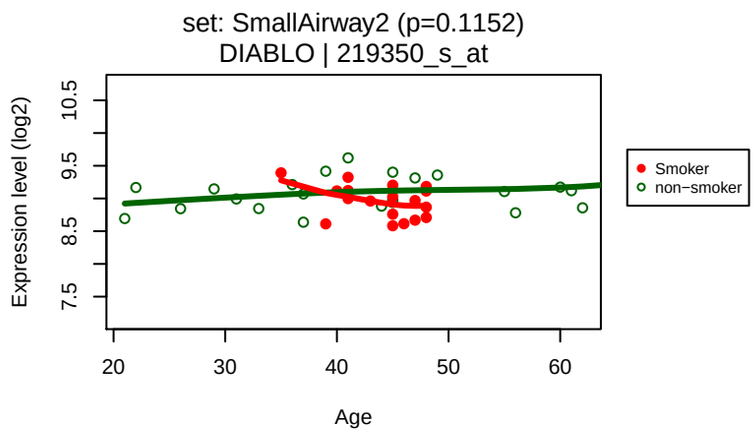
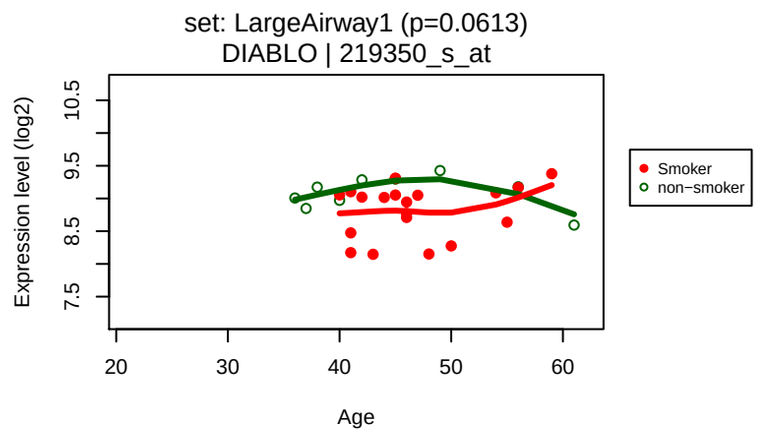

Gene expression of smokers and nonsmokers
gene: DIABLO | 219350_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0431
p=0.1152
p=0.0521
p=0.0613
p=0.9037
p=0.1944
p=0.3764
p=0.6568
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

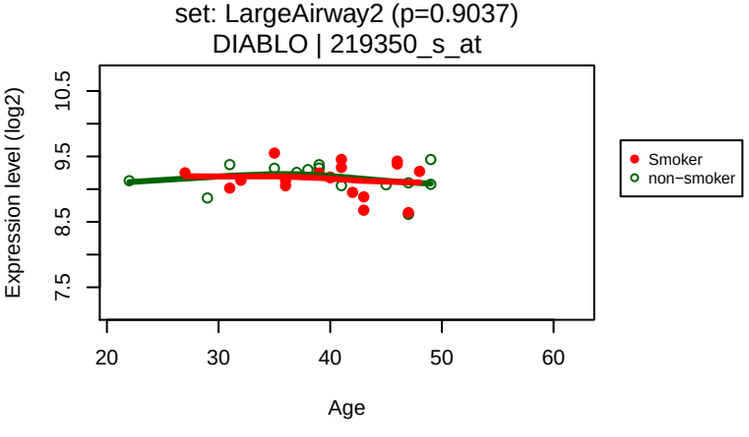

Gene expression ratio (smokers/nonsmokers)
gene: DIABLO | 219350_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0431
p=0.1152
p=0.0521
p=0.0613
p=0.9037
p=0.1944
p=0.3764
p=0.6568
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

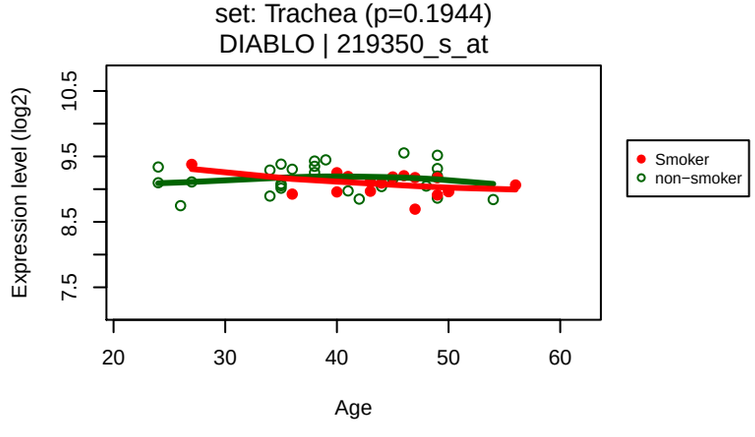

Gene expression ratio in Males
gene: DIABLO | 219350_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1161
p=0.5247
p=0.1665
p=0.8433
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DIABLO | 219350_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2867
p=0.3424
p=0.3941
p=0.0794
SmallAirway2
LargeAirway2
Trachea
Macrophage2

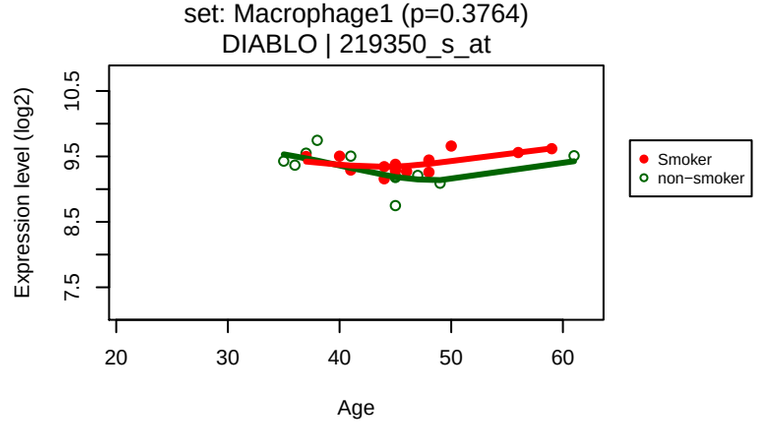

set: SmallAirway1 (p=0.0431)
DIABLO | 219350_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker

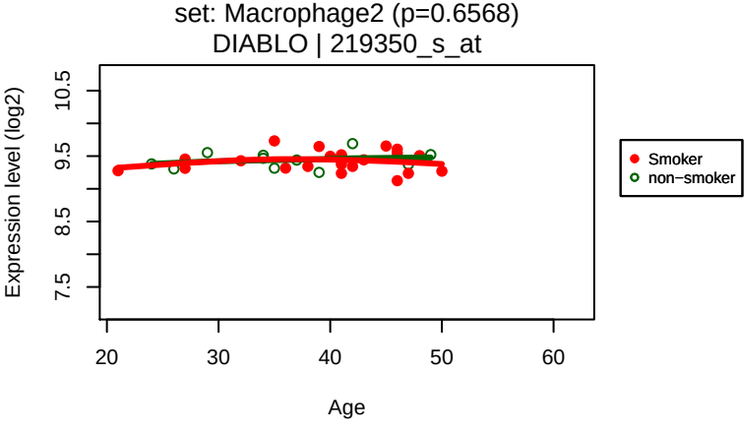

set: SmallAirway2 (p=0.1152)
DIABLO | 219350_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker

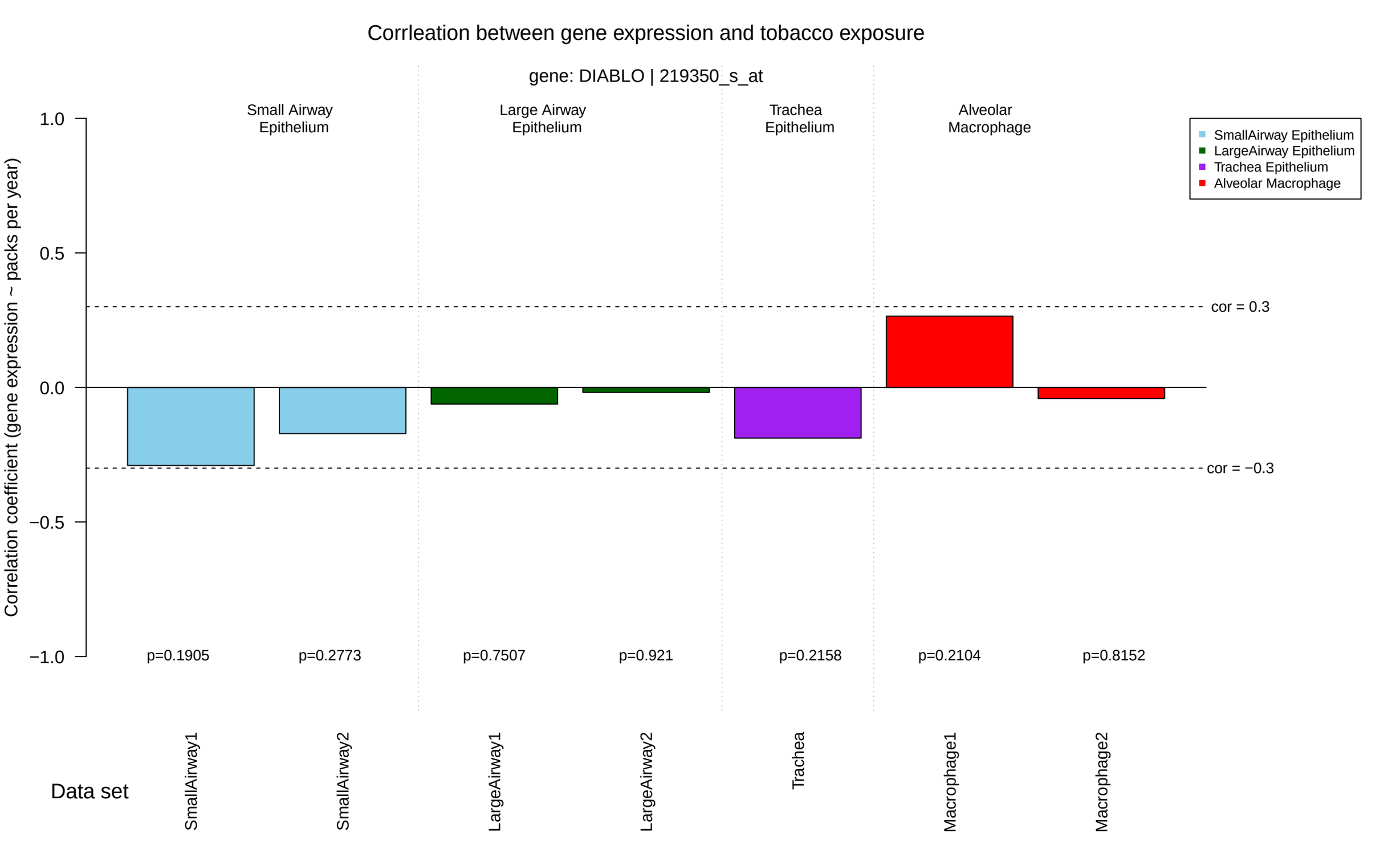

set: LargeAirway1 (p=0.0613)
DIABLO | 219350_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9037)
DIABLO | 219350_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.1944)
DIABLO | 219350_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3764)
DIABLO | 219350_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6568)
DIABLO | 219350_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DIABLO | 219350_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1905
p=0.2773
p=0.7507
p=0.921
p=0.2158
p=0.2104
p=0.8152
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2